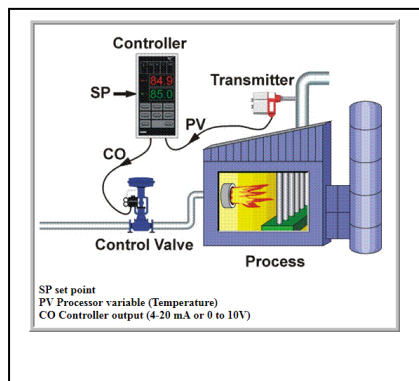




College:
Department:
Course:
Classroom and Time:

Instructor:
Office hours:
E-mail:
Cell Phone:

TA Name:
E-mail:
Office hours:
Office Location:

UIC COLLEGE OF ENGINEERING
UIC Department of Chemical Engineering
CHE 341 Chemical Process Control
EIB 271 M|W: 04:00 PM – 05:15 PM

Dr. Michael Caracotsios, EIB 244
M|Tu|W|Th: Stop by my office or schedule a ZOOM meeting
mcaracot@uic.edu
(630) 696-0934

Aanchal Vashisht
avash8@uic.edu
Th: 02:00PM - 04:00PM
EIB 228

COURSE OBJECTIVES

This course is designed to extend the student's understanding of the steady-state macroscopic and microscopic material and energy balances into the critical area of process dynamics. The major objective of the course is to emphasize and address the following issues pertinent to the safe and cost-effective operation of chemical plants. Given the process and operational targets, the course objectives would be to design a control system which is (a) stable, (b) has good performance characteristics, and (c) is robust. These design attributes require knowledge of dynamic behavior of processes (process modeling, solution of dynamic equations, characterization of dynamic behavior); control systems; stability and techniques for assessing it; performance criteria and how they are affected by controller parameters. By the end of the course the students should be able to:

- ✿ Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- ✿ Know how to develop empirical models from process data using the step response methodology
- ✿ Understand the theory of feedback, feed forward and cascade control and determine optimal controller settings for stable and robust performance.
- ✿ Know when to select and how to tune and troubleshoot feedback, cascade, feed forward and ratio controllers
- ✿ Apply available computer software (MATLAB, Excel or AthenaVisual Plus), or use Laplace transformation theory to solve process dynamic models and investigate their open and closed loop response to various external disturbances.

COURSE PREREQUISITES

The following are key requirements for a successful completion of this course.

- **Mathematics:** Differential and Integral Calculus
- **Statistics:** Linear and Nonlinear Regression
- Numerical Methods for Chemical Engineering
- **Programming Experience:** Excel VBA, MATLAB, C++
- **CHE Department Prerequisites:** CHE 301, CHE 311, CHE 312, CHE 313, **CHE 321** and **CHE 396 or CHE 433**

REQUIRED SOFTWARE

All students must download and install **ASPEN Plus v14.0** and the Microsoft EXCEL[32-bit Version] **CHE Calculator v13.0** following the instructions listed on the UIC Blackboard **Computational Tools** content area.

REQUIRED TEXTBOOK

Michael Caracotsios, *An Introduction to Chemical Process Control. Theory and Applications*. First Edition, 2026 will be distributed to all students free of charge via the UIC Blackboard in the **Course Textbook** content area of the CHE341 course.

RECOMMENDED TEXTBOOKS

1. D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, *Process Dynamics and Control*, Wiley, 2011.
2. J.B. Riggs and M. N. Karim, *Chemical and Bio-Process Control*, Ferret Publishing, Austin, Texas, 2016.
3. T.E. Marlin, *Process Control: Designing Processes and Control Systems for Dynamic Performance*, McGraw-Hill, 2000

CHE 341: COURSE OUTLINE

1. CHAPTER 1: REVIEW

- Mass and Energy Balances.
- Momentum and Mechanical Energy Balances.
- Entropy and Lost Work Balances.
- Calculus and Numerical Methods for Chemical Engineers

2. CHAPTER 2: INTRODUCTION TO CHEMICAL PROCESS CONTROL

- Control Degrees of Freedom
- Plant-Wide Process Control in Chemical Engineering

3. CHAPTER 3: MEASURING DEVICES AND FINAL CONTROL ELEMENTS

- Flow, Temperature and Pressure Measuring Devices
- Level, Composition and pH Measuring Devices
- Final Control Elements: Control Valves, Variable Speed-Drive Pumps and Dampers

4. CHAPTER 4: DYNAMIC MODELING AND CONTROL OF UNIT OPERATIONS

- Development of Dynamic Process Models
- Inventory Tanks and Static Mixers
- Reactors: Batch, CSTR and Plug Flow
- Process and Utility Heat Exchangers
- Distillation Columns and other Unit Operations

5. CHAPTER 5: LAPLACE TRANSFORMATIONS

- Laplace Transforms
- Linearization of Process Dynamic Models
- Derivation of Transfer Functions

6. CHAPTER 6: TRANSFER FUNCTIONS AND EMPIRICAL MODELS

- Process Identification from Plant Data
- First and Second Order Systems with or without Time Delays

7. CHAPTER 7: FEEDBACK CONTROL

- P, PI and PID Controllers: Parallel, Interactive and Non-interactive
- Block Diagrams and Routh-Hurwitz Stability Criterion
- Criteria for Controller Tuning
- PID Robustness and Bode Stability Analysis
- Guidelines and Troubleshooting Common Control Loops

8. CHAPTER 8: ADVANCED REGULATORY CONTROL (ARC)

- Feedforward and Cascade Control
- Direct Synthesis
- Gain Scheduling
- Ratio and Split-Range Control
- Principles of Internal Model Control (IMC) and Tuning methods
- Smith-Predictor for Delay Dominant Processes

9. CHAPTER 9: ADVANCED PROCESS CONTROL (APC)

- Interaction and Loop Pairing and Decoupling – Relative Gain Array
- Dual Composition Control
- Model Predictive Control of Linear Processes

CHE 341: COURSE GRADING AND OTHER POLICIES

Your final grade for the **CHE 341** Chemical Process Control course will be determined from (**max 10**) take home assignments, two midterms and one final examination, and (**max 10**) quiz/attendance records. The contribution of all these activities to the grade calculation formula is shown below:

1. Home Assignments: [40%]
2. Examinations: [40%]
3. Quizzes & Attendance: [20%]

The formula for the calculation of your total score is given by the equation below. The grades associated to a total score range are also given in the table below.

$$TS(\text{Total Score}) = 0.40\langle H \rangle + 0.40\langle E \rangle + 0.20\langle Q + A \rangle$$

Total Score	Grade	Symbol	Max Score	Meaning of Symbols
$85 \leq TS \leq 100$	A			
$70 \leq TS < 85$	B	$\langle H \rangle$	100 pts	Average Score for Home Assignments
$60 \leq TS < 70$	C	$\langle E \rangle$	100 pts	Average Score for Examinations
$50 \leq TS < 60$	D	$\langle Q + A \rangle$	100 pts	Average Score for Quizzes & Attendance

Note 1: Policy of Missed or Late Work:

Completion of all Assignments is mandatory. In the event that a student does not complete an Assignment by the designated time and date, the student will receive a score of zero points. The student may still be able to negotiate a better score by turning in the missed assignment at a time agreed upon with the Professor. If attendance is taken and a student is absent the student will receive a score of zero points, unless there is a reasonable explanation for the absence. Please email the Professor if you face an unexpected situation that may impede your attendance.

Note 2: Partial and Extra Credit Policy:

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Extra credit may be given at the discretion of the Professor. The extra credit will be based on the student's record of attendance, punctuality, participation in class, good work and ethics behavior, and reasonable scores on all assignments and examinations. The extra credit will be contingent on perfect attendance and completion of all home assignments.

Note 3: Academic Integrity Policy:

As a student and member of the UIC community, you are expected to adhere to the Community Standards of academic integrity, accountability, and respect. Please review the UIC Student Disciplinary Policy for additional information.

Note 4: E-mail Expectations:

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

Dr. Michael Caracotsios

CLINICAL ASSOCIATE PROFESSOR

CHEMICAL ENGINEERING DEPARTMENT

UNIVERSITY OF ILLINOIS CHICAGO

Email: mcaracot@uic.edu

MISCELLANEOUS

Blackboard Collaborate Help

By email: LTS@uic.edu

By phone: (312) 413-0003 Option #1

ACCOMMODATIONS

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: **"YOUR NAME: Requesting Religious Accommodation."**

CLASSROOM ENVIRONMENT

Inclusive Community

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

Name and Pronoun Use

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are [\[she/her; he/him; they/them\]](#). I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

Community Agreement/Classroom Conduct Policy

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.

- Approach discussion, challenges, and different perspectives as an opportunity to “think out loud,” learn something new, and understand the concepts or experiences that guide other people’s thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another’s privacy, and do not invite outsiders into our classroom.

Content Notices and Trigger Warnings

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

RESOURCES: Academic Success, Wellness, and Safety

We all need the help and the support of our UIC community. Please visit my **drop-in hours** for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

Academic Success

- [UIC Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#).
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)
- [First-at-LAS](#) Academic Success Program, focusing on LAS first-generation students.

Wellness

- **Counseling Services:** You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- **Campus Advocacy Network:** Under Title IX, you have the right to an education that is free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC’s Campus Advocacy Network at <http://can.uic.edu/>.

Safety

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.